

PERTH OBSERVATORY, Australia

WMO: 956100

Lat: 32.007S

Lon: 116.136E

Elev: 1263

StdP: 14.04

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.8	41.5	27.7	22.7	66.7	32.1	27.7	63.0	22.9	55.5	21.0	55.1	9.5	100	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.0	94.9	65.3	91.5	65.2	88.5	64.6	70.2	83.9	68.5	82.9	67.1	81.6	8.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	101.6	74.1	64.3	94.6	72.7	62.5	88.5	70.9	35.0	83.4	33.4	82.9	32.2	82.0	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.4	19.3	17.6	DB	36.9	101.3	1.6	2.9	35.7	103.4	34.8	105.1	33.9	106.7	32.7	108.8	
			WB	35.5	73.0	1.9	2.3	34.1	74.7	33.0	76.1	32.0	77.4	30.7	79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	72.6	72.7	69.6	63.9	58.0	53.4	51.7	52.6	54.4	58.9	64.7	69.3 (6)
	DBStd	9.24	5.92	5.73	6.27	5.46	4.31	3.48	3.34	3.65	4.48	5.39	6.24	6.73 (7)
	HDD50	56	0	0	0	0	1	9	21	15	9	1	0	0 (8)
	HDD65	2126	8	6	25	85	221	348	413	385	320	204	81	30 (9)
	CDD50	4343	701	635	606	417	249	111	73	95	140	276	441	599 (10)
	CDD65	939	243	221	167	52	4	0	0	1	14	73	164	164 (11)
	CDH74	9464	2635	2238	1483	319	20	0	0	1	16	158	782	1812 (12)
	CDH80	3870	1180	968	563	73	2	0	0	2	35	262	785	785 (13)
(14) Wind	WSAvg	9.6	10.4	10.5	9.9	9.2	8.9	9.2	8.9	9.0	9.2	9.7	10.0	10.1 (14)
(15) Precipitation	PrecAvg	38.3	0.6	0.6	0.8	2.0	4.9	7.7	7.8	5.8	3.9	2.2	1.3	0.5 (15)
	PrecMax	58.3	5.1	4.5	3.0	5.8	13.2	17.8	14.8	9.7	10.2	6.3	3.6	1.6 (16)
	PrecMin	24.8	0.0	0.0	0.0	0.0	0.8	1.2	3.5	0.7	0.5	0.0	0.1	0.0 (17)
	PrecStd	7.9	1.0	0.9	0.8	1.5	2.9	3.2	2.8	2.2	1.8	1.3	1.0	0.5 (18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.3	97.6	96.2	86.8	76.5	67.3	65.3	69.1	75.9	84.4	92.1	97.8 (19)
		MCWB	65.0	67.3	64.2	62.0	58.1	55.0	53.9	55.1	58.0	60.6	61.3	64.6 (20)
	2%	DB	94.4	93.1	90.5	81.6	72.1	64.6	62.1	65.0	70.5	78.0	87.0	93.0 (21)
		MCWB	65.7	66.3	65.0	61.4	57.5	54.3	53.9	54.4	56.2	59.6	61.6	64.3 (22)
	5%	DB	90.6	89.8	86.0	77.3	68.5	62.0	59.7	61.8	66.2	73.9	82.4	88.5 (23)
		MCWB	65.9	65.2	63.7	60.8	56.7	54.0	53.3	54.0	55.2	58.8	61.0	64.1 (24)
	10%	DB	87.0	86.3	82.2	73.7	65.5	59.9	57.6	59.3	62.5	69.8	78.2	84.0 (25)
		MCWB	65.0	64.7	63.4	59.9	56.5	53.8	52.7	53.3	54.1	57.5	60.2	63.0 (26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	72.5	70.1	66.9	63.7	59.8	59.0	59.7	60.9	64.1	66.7	69.3 (27)
		MCDB	85.9	85.7	82.1	75.2	68.9	62.4	60.8	62.4	69.8	77.0	81.1	86.0 (28)
	2%	WB	71.0	69.9	68.1	65.1	61.9	58.1	56.6	57.6	58.8	61.8	64.8	67.5 (29)
		MCDB	83.3	84.7	80.4	73.7	65.7	60.7	58.6	60.8	64.8	73.9	79.6	84.9 (30)
	5%	WB	68.8	68.1	66.5	63.5	60.1	56.6	55.1	55.8	57.2	60.1	63.2	65.9 (31)
		MCDB	82.8	83.4	79.3	72.0	64.3	59.1	57.5	59.4	62.7	70.2	76.6	83.2 (32)
	10%	WB	66.9	66.4	65.0	62.0	58.6	55.2	53.7	54.4	55.5	58.5	61.5	64.3 (33)
		MCDB	81.7	81.6	77.5	69.9	63.1	57.9	56.2	58.0	60.8	66.9	73.7	80.0 (34)
(35) Mean Daily Temperature Range	5% DB	MDBR	25.0	24.3	21.9	17.8	14.9	13.0	12.3	13.5	15.5	18.7	21.6	23.9 (35)
		MCDBR	30.3	28.9	27.1	22.9	19.8	16.0	15.0	17.4	21.9	26.1	29.2	30.3 (36)
		MCWBR	10.6	10.0	9.4	9.6	9.8	9.4	9.2	9.9	10.8	11.8	11.0	10.8 (37)
	5% WB	MCDBR	23.6	23.8	21.5	17.7	13.3	11.4	11.0	13.2	17.2	21.6	23.9	25.6 (38)
		MCWBR	10.2	10.1	8.8	8.9	8.5	8.6	8.6	9.1	9.8	10.7	10.6	10.6 (39)
(40) Clear-Sky Solar Irradiance	taub	0.323	0.323	0.309	0.301	0.293	0.286	0.286	0.291	0.311	0.311	0.313	0.312	
	taud	2.596	2.612	2.645	2.653	2.647	2.650	2.647	2.626	2.541	2.563	2.583	2.608	
	Ebn at Noon	322	316	310	297	284	278	284	296	305	317	324	327	
	Edh at Noon	33	31	29	25	23	21	22	25	31	33	33	33	
(44) All-Sky Solar Radiation	RadAvg	2594	2336	1928	1404	1050	858	909	1177	1568	2044	2442	2654	
(45)	RadStd	126	85	96	84	70	58	56	59	90	96	112	118	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page